

05

Fatigue Failure

Comprehensive Study Notes

Course Instructor

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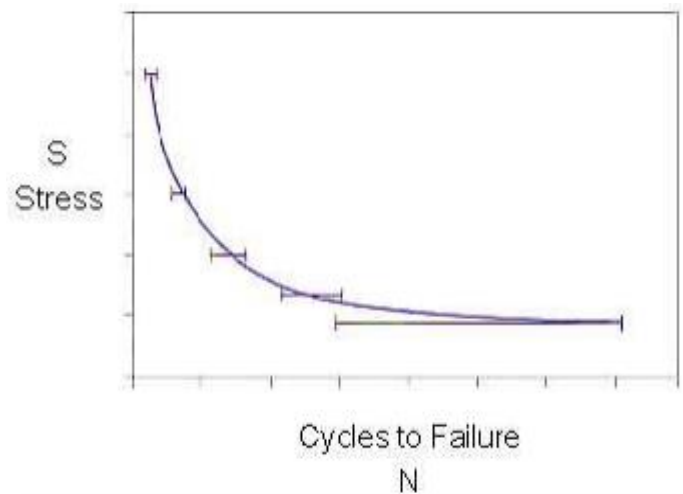
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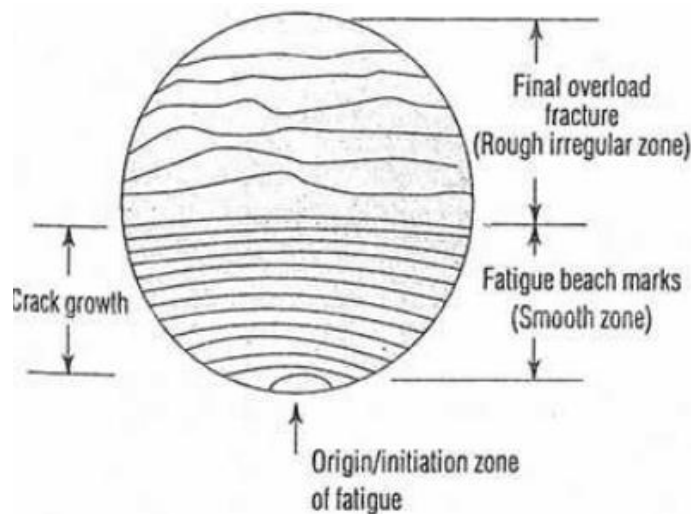
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FATIGUE FAILURE

Fatigue fracture results from the simultaneous action of repeated or fluctuating cyclic stress, tensile stress, and plastic strain. No fatigue crack starts or grows in the absence of any of these three active components. Cyclic stress initiates the crack and tensile stress produces the crack growth. There are many variables in service that influence the fatigue behavior or characteristic pattern. These include the magnitude and frequency of application of the fluctuating stress, the presence of a mean stress, temperature, environment, part size and shape, state of stress and residual stresses, surface finish, surface damages and microstructure. Schematic of S-N Curve, showing increase in fatigue life with decreasing stresses. The occurrence of fatigue may be considered as a three-stage process:



1. Initiation of the surface or sub-surface fatigue crack under a fluctuating load.
2. Crack propagation under tensile stress until the section thickness critically bears the imposed load.
3. Sudden fracture under overload.



On macro-examination the most characteristic feature of a fatigue-fracture is the presence of **beach/ conch shell** or **ratchet marks** which are centered around a crack origin (Fig. 4.8b). Normally the presence of these marks indicates a fatigue phenomenon, but the reverse need not be true, i.e., all the beach marks need not be indicative of fatigue. This is true in the case of high-speed steel brittle failure which exhibits a characteristic pattern of beach marks. **There is no evident macroscopic plastic deformation near the fracture / crack.** Fatigue-fracture surfaces are smooth textured near their origins carrying 'beach marks' which become coarser in texture as the crack grows and finally fails by ductile, shear or cleavage. Most fatigue cracks are transcrystalline without much branching, although intracrystalline fatigue is not uncommon. At high magnification (SEM or TEM), the most prominent fracture feature observed is **patches of fully spaced parallel marks, called fatigue striation** (Fig. 4.9). Striations orient themselves perpendicular to the direction of crack propagation and with

uniform loading increase in spacing as they progress. Each striation is the result of a single cycle of stress. Striation is confirmation of a fatigue. However, striations are not evident in high strength steels.

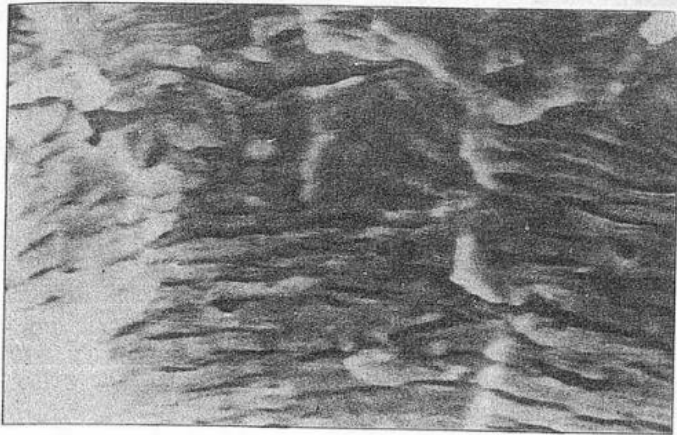


Fig. 4.9 Typical fatigue striations in a mild steel failure at high magnification



Fig. 4.8b Distinct beach marks due to fatigue, originating from the root radius zone of gear tooth (x 4)

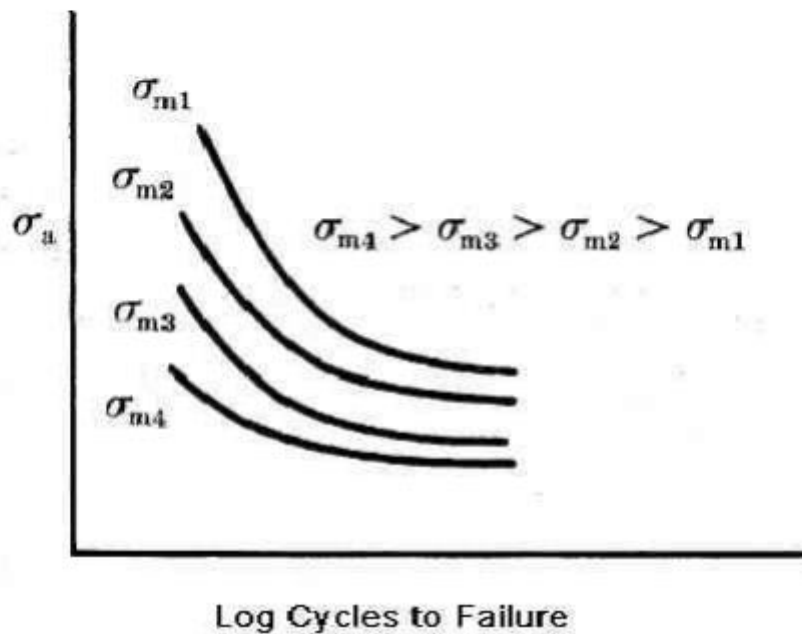
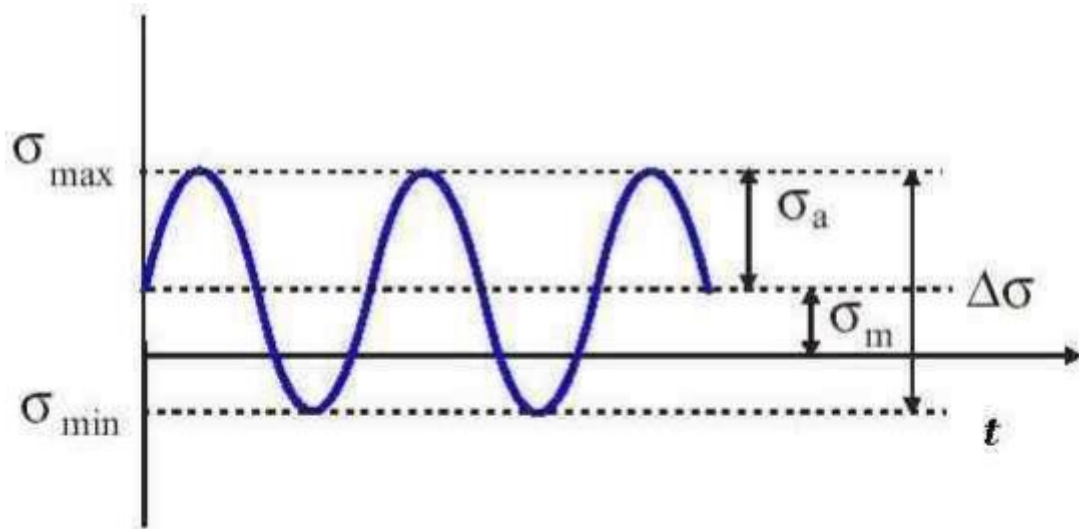
High Cycle (Above 10^6 Cycle)	Low Cycle (Below 10^5 Cycle)
• Smoother surface • Beach mark spacings are fine, uniform, and narrow • Arrest lines are clear and distinguishable under suitable magnification	• Rather coarse surface • Beach mark spacings are coarse, nonuniform and wide and far apart • Feeble arrest lines-sometimes difficult to distinguish between low fatigue and brittle fractures

In the presence of a severe stress concentration multiple crack origins are formed which eventually unite to form a single crack front. When the stress concentration on the surface is absent cracks propagate more readily near the center of a section than at the surface. While there is a stress concentrating 'notch', W-shaped crack fronts are observed. For a given material, the zone of the final fast f' fracture increases with increasing applied load. On rotating or bending the final fracture region is often rotated or offset towards the origin in the direction opposite to the direction of rotation. Also, with increasing nominal stress, the final fracture moves towards the center.

Stress Ratio:

- The most commonly used stress ratio is R , the ratio of the minimum stress to the maximum stress (S_{min}/S_{max}).
- If the stresses are fully reversed, then $R = -1$.
- If the stresses are partially reversed, $R =$ a negative number less than 1.
- If the stress is cycled between a maximum stress and no load, $R =$ zero.
- If the stress is cycled between two tensile stresses, $R =$ a positive number less than 1.

Variations in the stress ratios can significantly affect fatigue life. The presence of a mean stress component has a substantial effect on fatigue failure. When a tensile mean stress is added to the alternating stresses, a component will fail at lower alternating stress than it does under a fully reversed stress.



As the mean stress is increased, fatigue life decreases. This occurs because the stress applies is higher.

Fatigue Types:

1. Bending Fatigue

Fatigue can result from three types of bending loads, viz. unidirectional (one way), reversal (two way) and rotating. When a shaft is subjected to a bending stress, the convex surface experiences the tensile component of the stress while the concave surface remains compressed.

a) Unidirectional Bending Fatigue

Beach marks, indicative of a fatigue crack, show a single origin (at the point indicated by the arrow in Fig. 4.10 (a)). The crack front with its beach marks shows symmetry with respect to the origin and retains a concave form throughout. Both the single origin and the smallness of the final fracture zone in Fig. 4.10 (a) are suggestive of a low nominal stress. The larger final fracture zone in Fig. 4.10 (b), indicates a fatigue failure under a higher (moderate) nominal stress. However, fatigue cracks under high nominal stress seem to originate as several individual cracks that ultimately merge to form a single crack front (Fig. 4.10c). Usually characteristic 'ratchet' marks are present between crack origins. Figures 4.10 (d) (e) and (f) indicate typical beach marks, under moderate stress concentration, provided by a change in section in a uniformly loaded shaft. With a nominal stress, the crack front changes from concave to convex prior to rupture (Fig. 4.10(d)) At higher nominal stresses, the crack front flattens as Fig. 4.10(e) and (t). With severe stress concentration produced by a change in section in a uniformly loaded shaft, beach marks show a typical pattern as in Fig. 4.10(g), (h) and (i). Multiple crack origins, high nominal stress and unidirectional bending producing the fatigue beach marks are shown in Fig. 4.10(i).

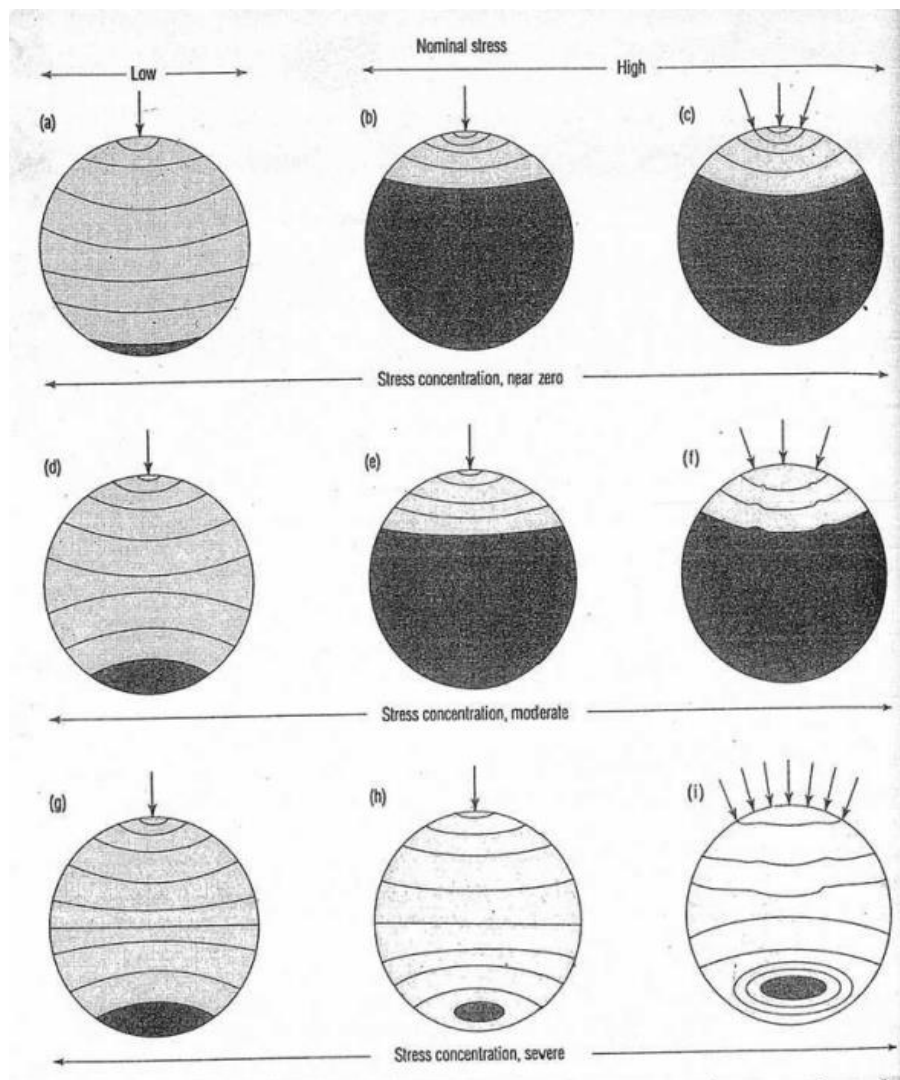
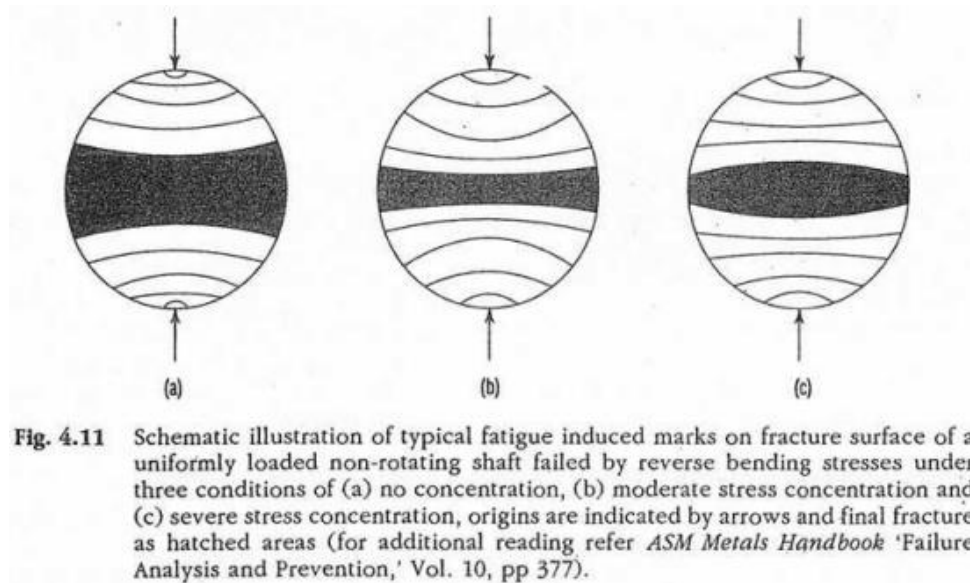


Fig. 4.10 Schematic illustration of typical fatigue induced marks on fracture surfaces of a uniformly loaded shaft, produced from a single origin when subjected to unidirectional bending at low and high nominal stresses and those features obtained from multiple origins at high nominal stress. Origins are indicated by arrows and final fractures as hatched areas, (for additional reading ref. *ASM Metals Hand Book, Failure Analysis and Prevention*, Vol. 10, page 376)

b) Reverse Bending (Flexural) Fatigue Failure

If the applied bending moment is reversing and maintains the same magnitude in either direction, two cracks of almost equal length are identically developed from the origins located diametrically opposite to each other and usually in the same transverse plane. However, two cracks differ in length in case the bending moment is found greater in one direction than in the other. Figure 4.11 shows typical fatigue marks produced on the fracture surface of a uniformly loaded non-rotating shaft subjected to reversal bending stresses. The pattern shown in Fig. 4.11 (a) is typically that of a single diameter shaft with no stress concentration, indicating equal bending moment in both directions. Fig. 4.11 (b) shows the typical pattern of beach marks due to a moderate stress concentration imposed by a large radius fillet at a change in the shaft diameter. Figure 4.11 (c) gives the typical pattern due to a severe stress concentration by a small radius fillet at a change in the shaft diameter.



- These are transverse loading (tension-compression)
- The catastrophic zone is at the center of the shaft, however, can be shifted towards one side.
- Origins formed on each side.
- Each origin creates its own clam shells and can have different flatness.

Figure a:

- High applied stresses.
- LCF compared to b
- No flipping clam shells

Figure b:

- Low stress concentration (more concave shape) and low applied stresses (small catastrophic zone).

Figure c:

- Direction of clamshells flipped in the middle with moderate-high stress concentrations
- Low applied stresses, therefore small catastrophic zone (bigger than b).

c) Rotary Bending Fatigue

In contrast to a stationary shaft, where tensile stress is only confined to a portion of the periphery, the rotating shaft sustains a tensile stress at every point on the periphery, with the asymmetrical development of the crack front from a single Origin. In such a case, a crack may be initiated at any point on the periphery [Fig. 4.12(b) and (d)]. The crack front which tends to extend in a direction opposite to that of rotation, usually swings around 15° or more [Fig. 4.12(a) and (c)].

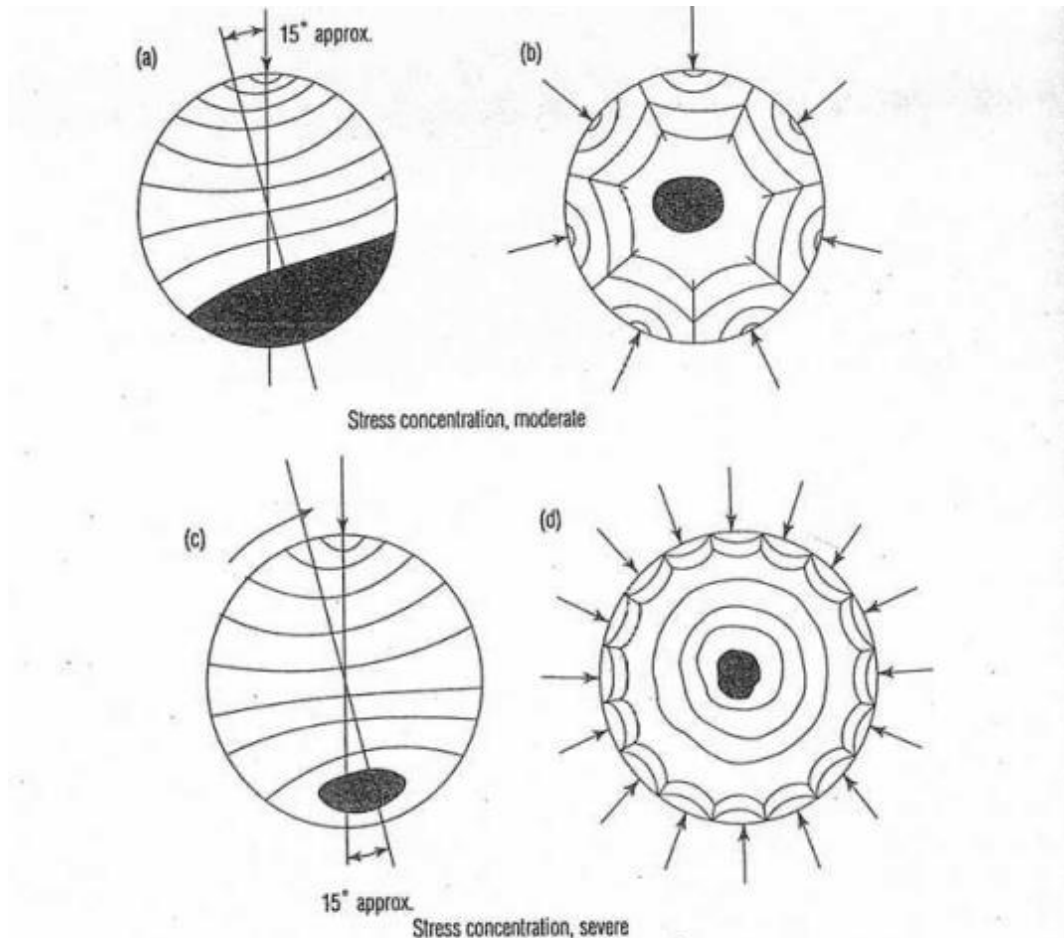


Fig. 4.12 Schematic illustration of fatigue induced marks on fracture surfaces of a uniformly loaded rotating shaft, originating from single and multiple origins under moderate and severe stress concentrations. The shaft rotation is in the clockwise direction. The origins are indicated by arrows and final fractures as hatched areas, (for additional reading ref. *ASM Metals Handbook, Failure Analysis and Prevention*, Vol. 10, page 377)

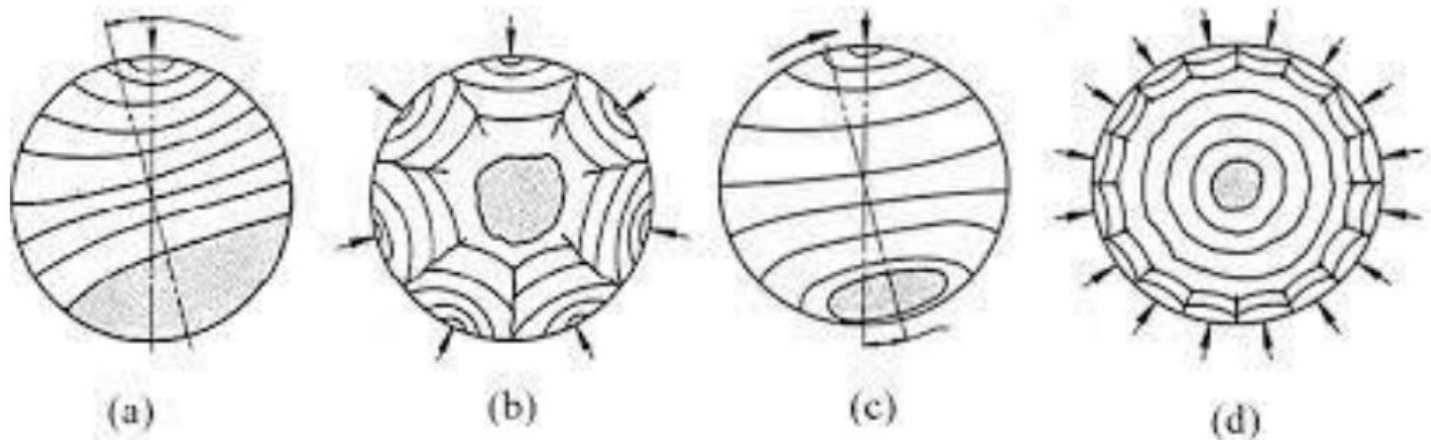
Rotational fatigue causes origins all around.

Figure b:

- Clam shells restricted indicating low stress concentration

Figure d:

- Flipping clam shells indicates high stress concentration.
- Clam shells merged together indicating severe stress concentration. (beysama3 fel genab)



2. Torsion Fatigue Fracture

Fatigue cracks resulting from torsional stresses show the same beach marks and ridges as those produced by bending stresses (Fig. 4.13). Under torsional loading of a shaft, the maximum local tensile stresses are at 45° to the axis of the shaft. Under a fluctuating torsional load, fatigue cracks may develop normal to the tensile stresses. Under alternating (reversed) torsional loading, two sets of fatigue cracks perpendicular to each other may develop.

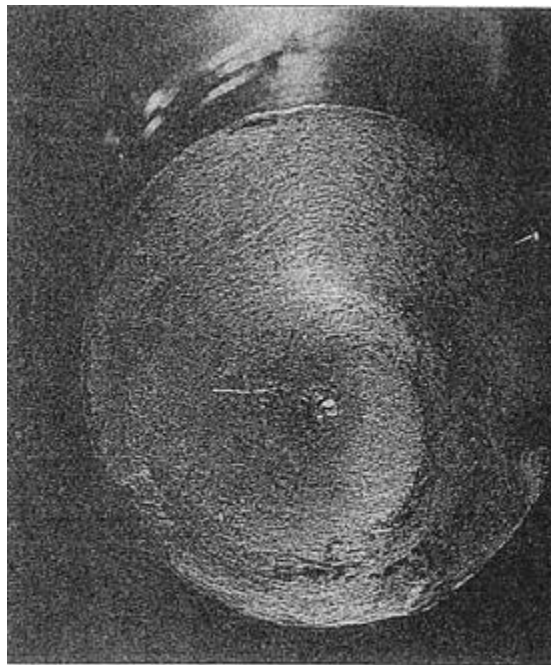


Fig. 4.13 Fracture surface of a shaft that failed by torsional fatigue ($\times 3$)

3. Contact Fatigue Fracture

This type of failure is commonly observed in parts (such as bearing) involving rolling or rolling and sliding operations against each other under high contact pressure and cyclic loading. The progressive development of cyclic stresses, at or near the contact surface during operation results in a gradual deterioration of the material by damage mechanisms leading to initiation and propagation of fatigue cracks. In some failures, small pits are initially observed on the polished contact surfaces before the spalling starts. Spalling is characterized by surface deteriorations like pits, crater-like cavities in the highly stressed surfaces of raceways or rolling elements (Fig. 4.14). The irregular cavities are sharp edges with steep walls and flat bottoms. Spalling is also recognized as a severe form of flaking.

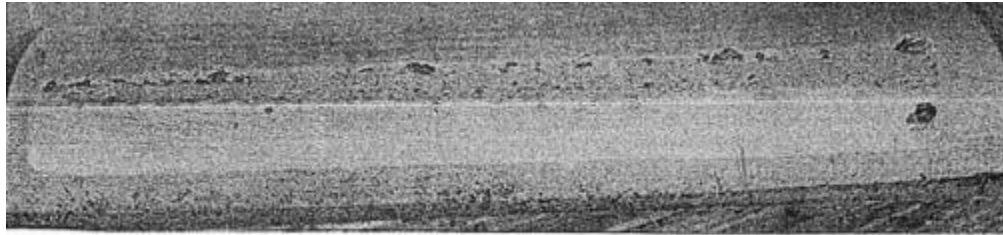


Fig. 4.14 Spalled tooth surface of a failed gear in design testing

Generally, spalling (Fig. 4.15) in a roll is attributed to initiation of a subsurface fatigue crack of undetermined origin or is a result of surface induced microscopic cracking. Spalling may originate below the surface as a result of localized stress concentration.

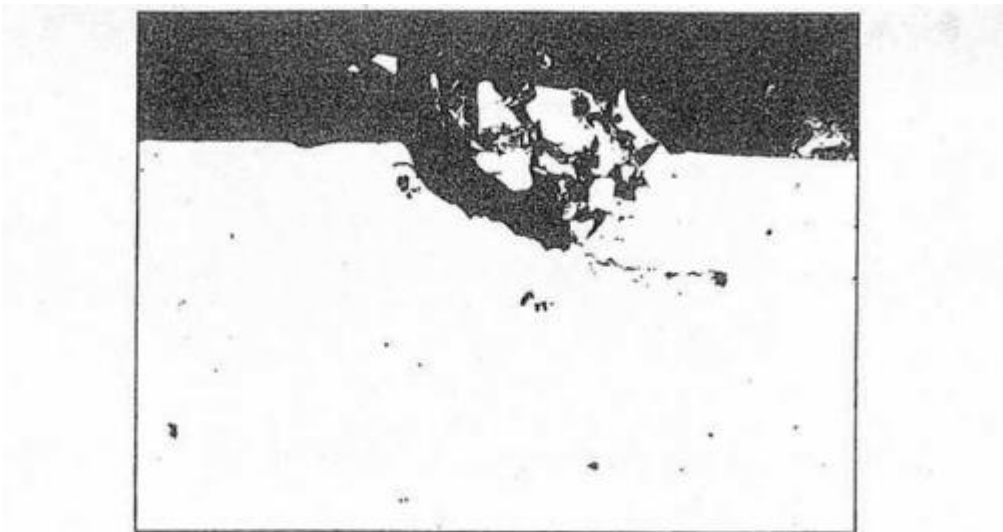


Fig. 4.15 A typical spalling characteristic phenomenon induced by a sub-surface fatigue crack in the section of a case-carburised steel ($\times 400$)

4. Brinelling

Fatigue failures due to brinelling effect are commonly noticed in a stationary bearing or engine component under excessive pressure or impact loading. Permanent deformations, like hollows or dents, are caused by plastic flow of metal and identified as indentations corresponding to ball spacing in the ball bearing component. These characteristic features differentiate brinelling from false brinelling (produced by vibrations or oscillation) and dents made by debris (in which case dents are spaced randomly).

5. Thermal Fatigue Failure

Thermal fatigue failure is the result of repeated thermal cycling which generates repeated stress gradients within a free component, or stress cycling within a component constrained to fixed dimensions. Thermal fatigue fractures are characterized by surface crazing formed by multiple initiation sites that join randomly by edge sliding to form the main crack as shown in Fig. 4.16. Other features to identify thermal fatigue are:

- Fractures are planar and transverse with no visible plastic deformation
- Fracture is mostly transgranular
- Oxidized fracture surfaces and oxide wedge filled cracks further characterize thermal fatigue failures

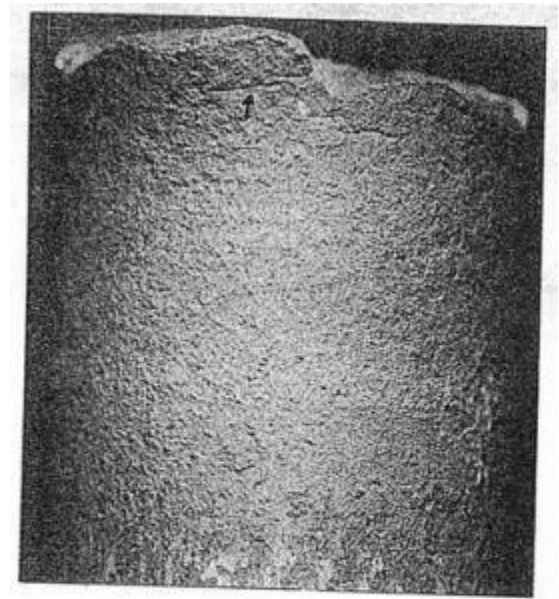


Fig. 4.16 Shows multiple initiation zones of thermal fatigue on the surface of an engine inlet valve (x6)

6. Fretting Corrosion Fatigue

Generation of oxide debris at the interface of two surfaces in contact, when subjected to relative oscillatory motion by virtue of adhesion, indicates the phenomenon of fretting corrosion. The rubbing action removes the protective film on a metallic surface and the small particles that are created then oxidize and act as abrasives inducing fretting, pitting. Fretting destroys the dimensional accuracy of closely fitted parts, ruins bearing - surfaces (particularly ball and roller bearings) and increases the susceptibility to fatigue failure. When fatigue failure results due to initiation and propagation of fatigue cracks from such fretted areas, it is termed fretting fatigue. Characteristics:

- Surface discoloration due to oxidation, most commonly noticed in ferrous materials such as reddish-brown color and blackish color in aluminum alloys
- Formation of debris or welded zones and deep pits at the contacting surfaces (Fig. 4.24)
- Fatigue crack originating at the fretted zone
- Other fracture features are similar to fatigue failure

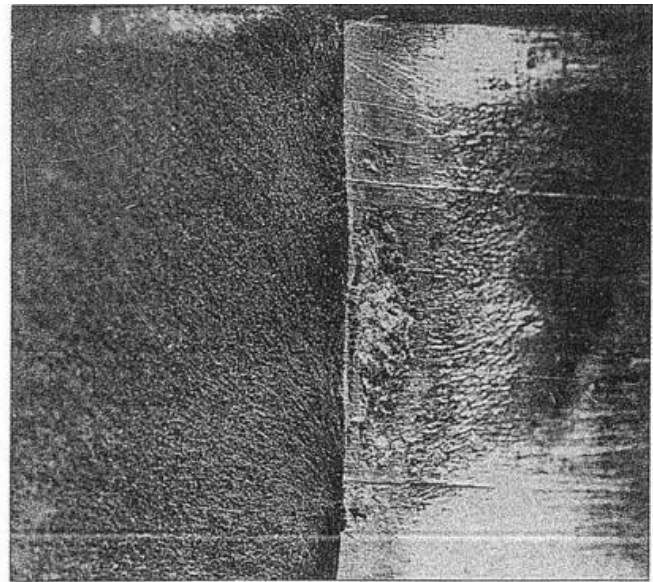


Fig. 4.24 A typical fretting fatigue fracture showing the origin (marked by arrows) of fatigue on the fretted zone in a failed rotating steel part (x 3)

7. Corrosion Fatigue

It occurs due to the combined action of corrosion and cyclic stress on metal parts. The corrosion fatigue mechanism involves the formation of extrusions and intrusions at the surface, destroying the protective surface layer at these points and allowing electrochemical attack to form microscopically small discontinuities. These discontinuities act as a notch for fatigue fracture to occur under the influence of continuous chemical reaction. Fatigue life of a dynamically loaded parts may be reduced by 50% in a moist or corrosive environment than in dry air. The process of failure by corrosion fatigue is composed of two stages. In the first stage the combined action of corrosion and cyclic stresses damage the metal by pitting and initiate a crack. The second stage is essentially a fatigue stage in which fatigue crack propagation is accelerated by corrosion and is controlled primarily by stress corrosion effects and the physical properties of the metal. Characteristics:

- Surface of the metal is generally pitted, and cracks are initiated at the bottom of the pit (Fig. 4.25)
- Cracks are predominantly transcrystalline and progress in the direction normal to the operating stress
- Cracks are intracrystalline in ferritic steels (particularly 13% chromium steel) depending on the stress intensity
- Several cracks initiate and propagate simultaneously
- Fracture features are identical to those of fatigue failure with its origin at the corrosion pit. However, these features are sometimes obliterated to some extent by corrosion

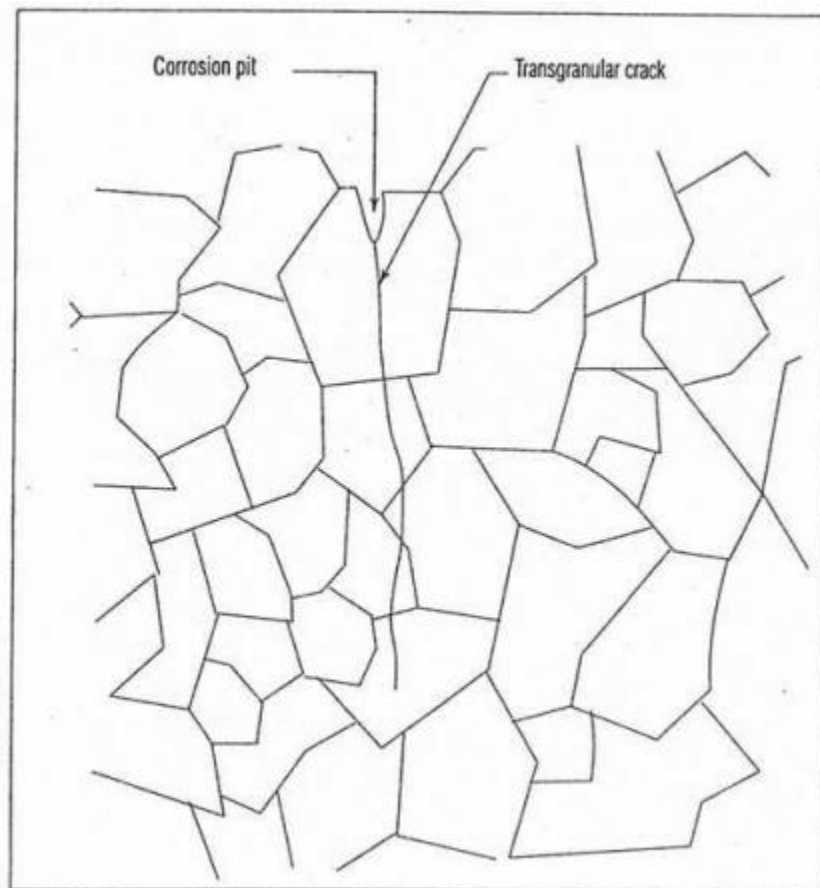


Fig. 4.25 Schematic representation of a corrosion fatigue cracking

Preventing Fatigue Failure:

The most effective method of improving fatigue performance is improvements in design:

- Eliminate or reduce stress raisers by streamlining the part (e.g., sharp corners, inclusions, etc.)
- Avoid sharp surface tears resulting from punching, stamping, shearing, or other processes
- Prevent the development of surface discontinuities during processing.
- Reduce or eliminate tensile residual stresses caused by manufacturing.
- Improve the details of fabrication and fastening procedures
- Surface hardening either by case hardening of steels, shot peening of other alloys or use Erosion resistant coatings. e.g., Thermal brier coatings

Fatigue Failure Mechanism:

The most common cause of damage to engineering components is fatigue, because of dynamic rather than static loading, particularly in aeronautical applications. It is implied that the maximum load or stress to cause fatigue failure is much less than the static breaking load. This type of fracture usually occurs in the components that are subjected to repeated applications of alternating stress.

The tensile stress, whether applied or residual, is the major component of fatigue failure and is generally much lower than would be required for static fracture. After some critical number of stress cycles, a crack will form in the region of highest stress concentration. With continued cycling, the crack will grow in length and in a direction perpendicular to the applied tensile stress. The initiation and propagation of a fatigue failure is a very complicated process, both occurring as a plastic deformation. Unlike ductile fracture, fatigue cracking usually does not show any deformation or necking in the region of failure. Therefore, no prior indication of the onset of fatigue cracking can be detected visually. Normally fine cracks are difficult to be detected until they grow to macroscopic sizes leading to partial or complete rupture.

The basic method of presenting engineering fatigue data is usually in the form of S-N curves (i.e., fatigue curves with stress against the number of cycles to failure). Figure 6.1 illustrates a typical S-N curve from rotating beam tests.

It is seen that the S-N curve is concerned chiefly with fatigue failure at a high er number of cycles (i.e., $N > 10^5$ cycles). Under these conditions the stress on a gross scale is elastic. At higher stresses, the fatigue life is progressively decreased. For low cycle fatigue which is concerned with $N < 10^4$ cycles, tests are conducted with controlled cycles of elastic plus plastic strain instead of controlled stress cycles²³. As strain amplitude increases, N cannot be sustained at higher values. Hence it drops.

Low cycle fatigue failure is taken into consideration in the design of nuclear pressure vessels, steam turbines and other types of power machinery, whereas high cycle fatigue is concerned with dynamic structures. Low cycle fatigue often occurs where repeated, stresses are of thermal origin, due to thermal stresses produced by thermal expansion of the material. Hence, this type of fatigue is due to strain rather than stress cycling. The usual method of presenting low cycle fatigue data is by plotting the plastic strain versus number of cycles on a log-log scale in the form of a straight line.

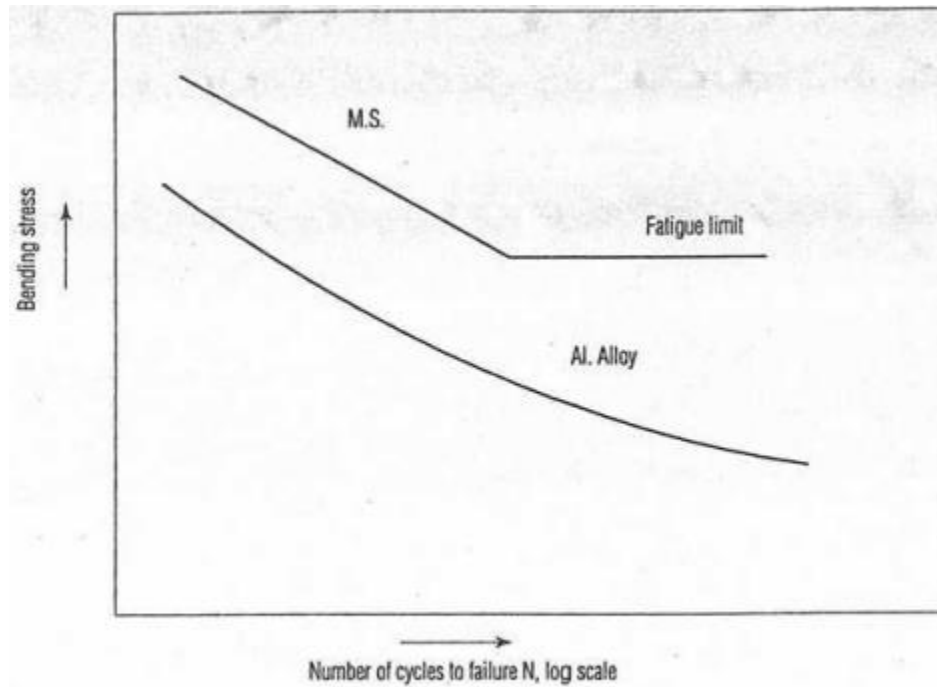


Fig. 6.1 Illustration of relationship between stress amplitude and life in cycles

Four Steps of Fatigue:

1. micro crack nucleation
2. micro crack propagation (Stage I)
3. macro crack propagation (stage II)
4. final rupture

Role of Stress Raisers in Fatigue:

A stress raiser is an irregularity from a geometrical point of view either in shape/section or surface causing a peak in stress at or near the irregularity and which is higher than the average stress in the surrounding areas. Stress raisers can be mechanical or metallurgical. Mechanical stress raisers are actual discontinuities or separation in a product (viz. fold, inclusion, lamination, crack, etc.). Metallurgical stress raisers arise from the localized differences in properties such as soft and hard spots, decarburization, grain flow variation, chemical segregation, banding, etc.

Source/Origin of Stress Raisers:

Since a large part of fatigue failure analysis involves detection and identification of stress raisers at the fatigue initiation site, four categories of sources are broadly identified:

1. Resulting from design weaknesses or defects (i.e., holes, corners, keyways, threads)
2. Carried over from metal refining processes (i.e., pipe, inclusion, segregation)
3. Caused during the fabrication process (i.e., seams, laps, tears, cracks)
4. Induced as a result of operation/environment (i.e., corrosion, irradiation, foreign objects, and damages)

Fatigue Crack under Alternating Loading:

Microscopic and macroscopic aspects of fatigue crack nucleation and growth.

The microscopic aspect of fatigue involves the phenomenon of crack nucleation and growth with the help of theories of micro-plasticity, whereas the macroscopic aspect deals with the growth of a pre-existing macro-crack or a crack like defect with the help of macro-plasticity. A free surface normally constitutes the preferential area for the initiation of a fatigue crack. Alternating stresses produce slip bands. In fatigue some grains will show slip lines without any evidence of slip in others. Figure 6.2 shows two types of surfaces deformation²⁴ namely, extrusions (material projects from the surface) and intrusions (where shallow channels are formed).

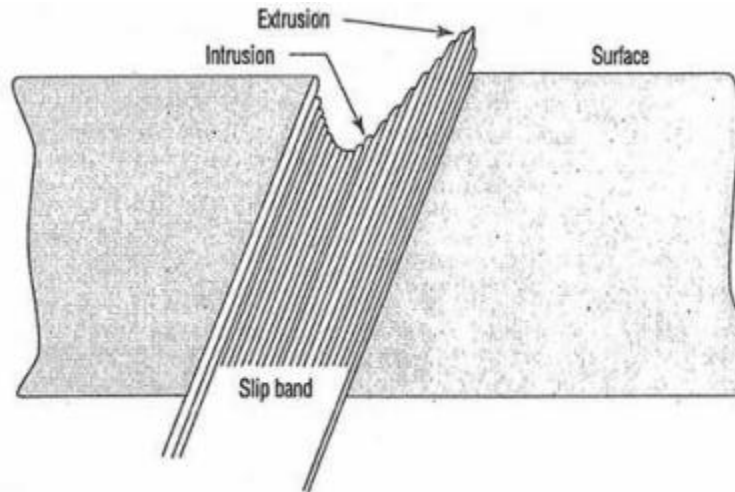


Fig. 6.2 Surface deformation processes of extrusions and intrusions in metallic materials subjected to alternating loading (for additional reading ref. J.C. Gross Kreutz, ASTM STP 495, 1971)

The slip lines are formed during the first few thousand cycles of stresses and the successive cycles produce additional slip bands. There are generally several slip bands which are more persistent than the rest and which will remain visible, while others are polished away. These persistent slip bands constitute the embryonic stage for fatigue cracking.

EM micrographs of intrusions and extrusions have indicated that the process is associated with the formation of micro notches and micro-cracks. Once formed, the fatigue crack tends to propagate at an angle of approximately 45° to the tensile stress (stage I of the crack propagation). The surface where the latter takes a direction perpendicular to the maximum applied stress is known as the stage II of crack propagation (Fig. 6.3). The 45° crack is easily identified as the narrow initiation zone on the fracture surface after failure in most of the alloys, except in high strength nickel and cobalt based alloys²⁴ where large areas of fracture surface themselves consist of this type of crack. The rate of crack propagation in stage I is very low and is of the order of a few atom diameters per cycle compared to rates a few microns per cycle in stage II.

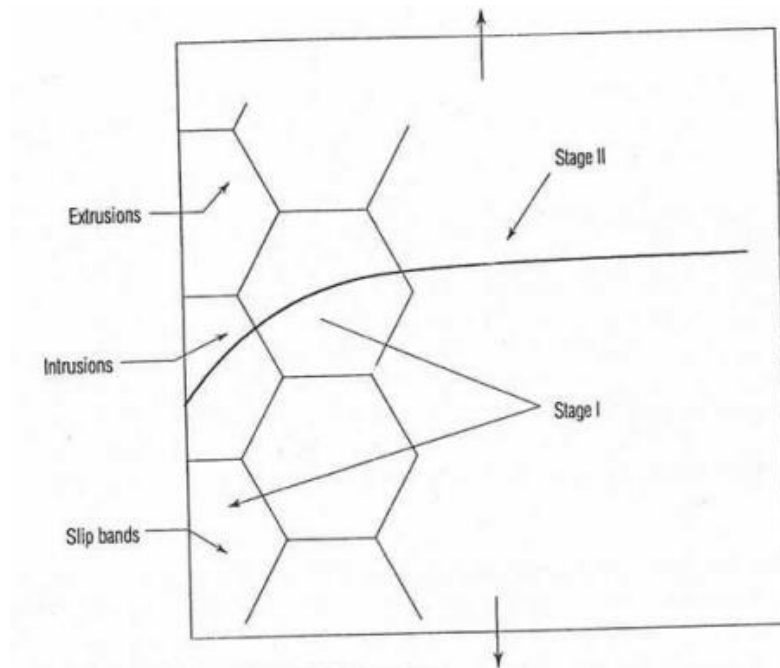


Fig. 6.3 Representing fatigue crack propagation in Stage I and Stage II (for additional reading ref. C. Laird, ASTM STP 415-1967)

An important microscopic feature which is unique to fatigue deformation is the formation of ridges and grooves on both the intrusion and extrusion slip bands. A mechanism of formation at intrusion and extrusion is shown schematically in Fig. 6.4.

During the tensile part of the stress cycle, both slip systems operate sequentially to produce surface stress. When the stress becomes compressive, intrusion forms by slip on the first system and extrusion forms when the other slip system operates. It is known that a fatigue failure can also occur in a perfectly smooth polished specimen without any apparent presence of stress concentration. But the crack initiation process is considerably influenced by irregularities on the surface, such as, inclusions at the free surface, foreign particles at the sub-surface or cracks formed by shot peening, scratches and surface damage caused by fretting or other mechanical means.

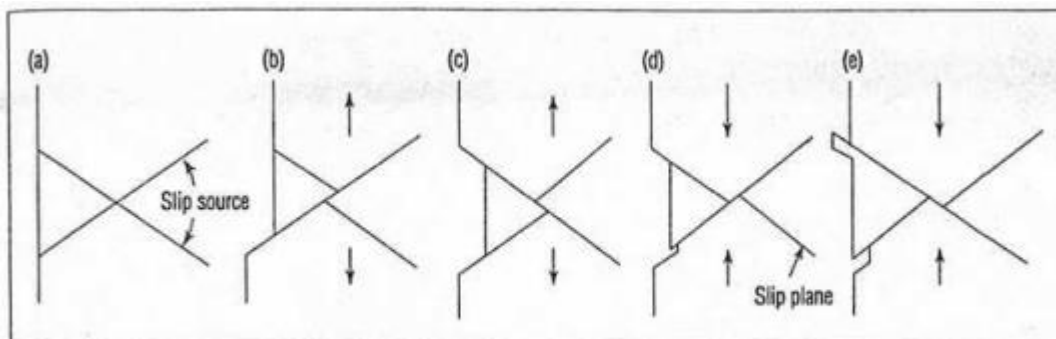


Fig. 6.4 Illustration for the mechanism of formation of slip band, extrusion and intrusion

Fatigue Crack Propagation:

After crack initiation in Stage I, Stage II starts with the propagation of crack at right angles to the direction of tensile stress till fracture takes place due to overloading (Stage III of crack propagation). During Stage II, the crack advances a certain distance with each load cycle and in the process a series of striations are produced on the fracture surface. The crack propagation in Stage II shows a characteristic propagation pattern at microscopic level and is indicated on the fatigue fracture surface. They reveal the extent of fatigue caused by the crack front from the transition to the final catastrophic fracture. The variations in the beach mark pattern (which are lying parallel to the fatigue striations) form due to changes in loading and/or in the crack tip environment.

Fatigue cracks propagate commonly in a ductile manner. Each load cycle produces an intensive plastic deformation around the crack tip and the cracking process takes place predominantly in a transcrystalline manner. In the case of polycrystalline microstructures, the crack front is sub-divided locally on to a number of separate planes²⁴ (Fig. 6.6), resulting in the formation of the microscopically visible structure of many parallel crack paths. The profiles of the individual striations depend on the material and the loading condition.

The formation of individual striation occurs as a result of successive opening and closing of the crack flanks. In the process, the crack advances during opening, whereas in closing, the plastically deformed material at the crack tip gets compressed. As a result, a fold is produced which subsequently stands out in the form of a striation. The variation in striation profile depends on the intensity of loading and the ease of slip formation at the crack tip.

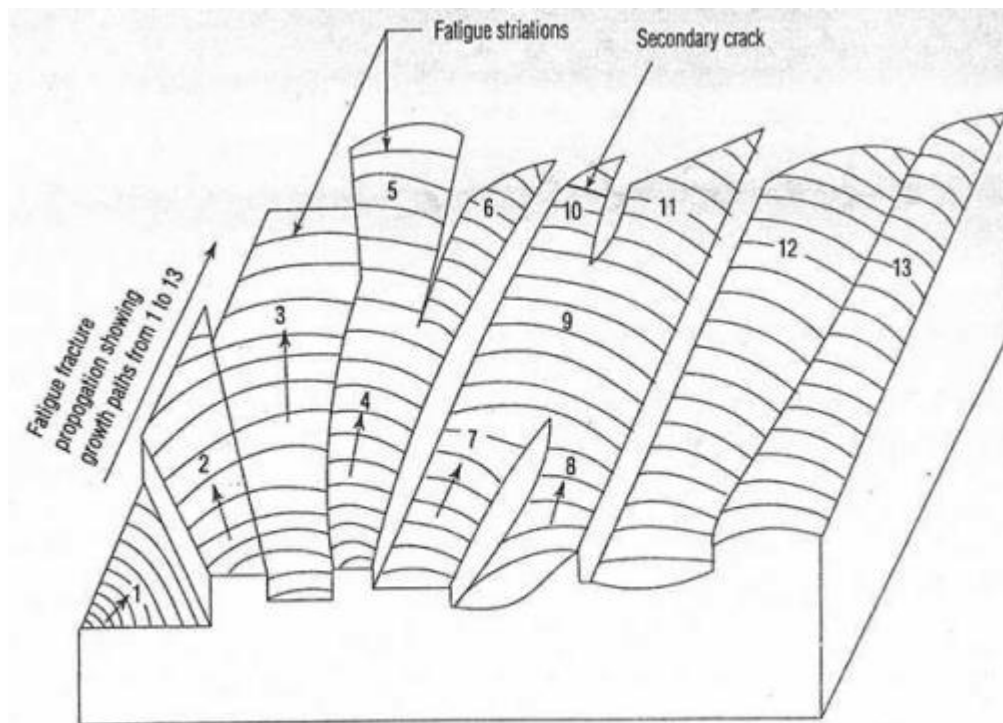
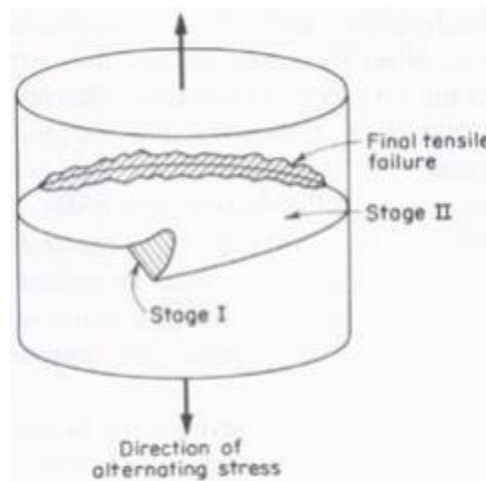


Fig. 6.6 Schematic representation of fatigue growth path in different planes
(For additional reading ref. C.D. Beachem Trans ASM 60, Philadelphia, USA 1967 S 325)

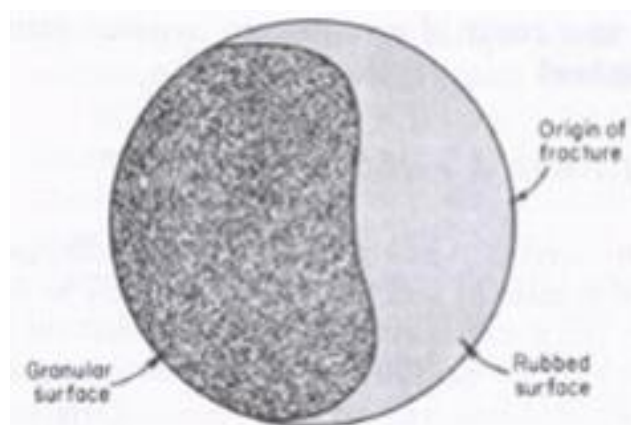
Crack Initiation and Propagation:

Failure of a material due to fatigue may be viewed on a microscopic level in three steps:

1. **Crack Initiation:** The initial crack occurs in this stage. The crack may be caused by surface scratches caused by handling, or tooling of the material; threads (as in a screw or bolt); slip bands or dislocations intersecting the surface as a result of previous cyclic loading or work hardening.
2. **Crack Propagation:** The crack continues to grow during this stage as a result of continuously applied stresses
3. **Failure:** Failure occurs when the material that has not been affected by the crack cannot withstand the applied stress. This stage happens very quickly.

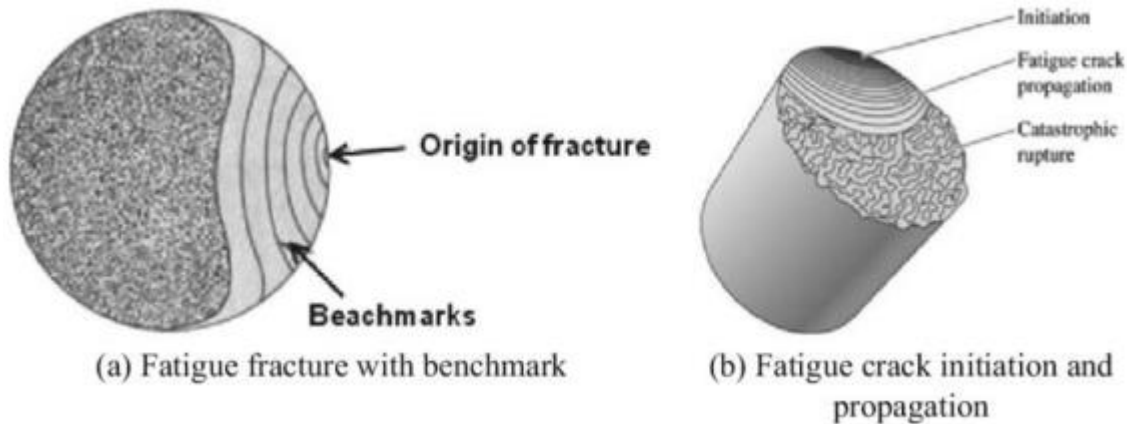


One can determine that a material failed by fatigue by examining the fracture sight. A fatigue fracture will have two distinct regions; One being smooth or burnished as a result of the rubbing of the bottom and top of the crack (steps 1 & 2); The second is granular, due to the rapid failure of the material.



Other features of a fatigue fracture are Beachmarks and Striations:

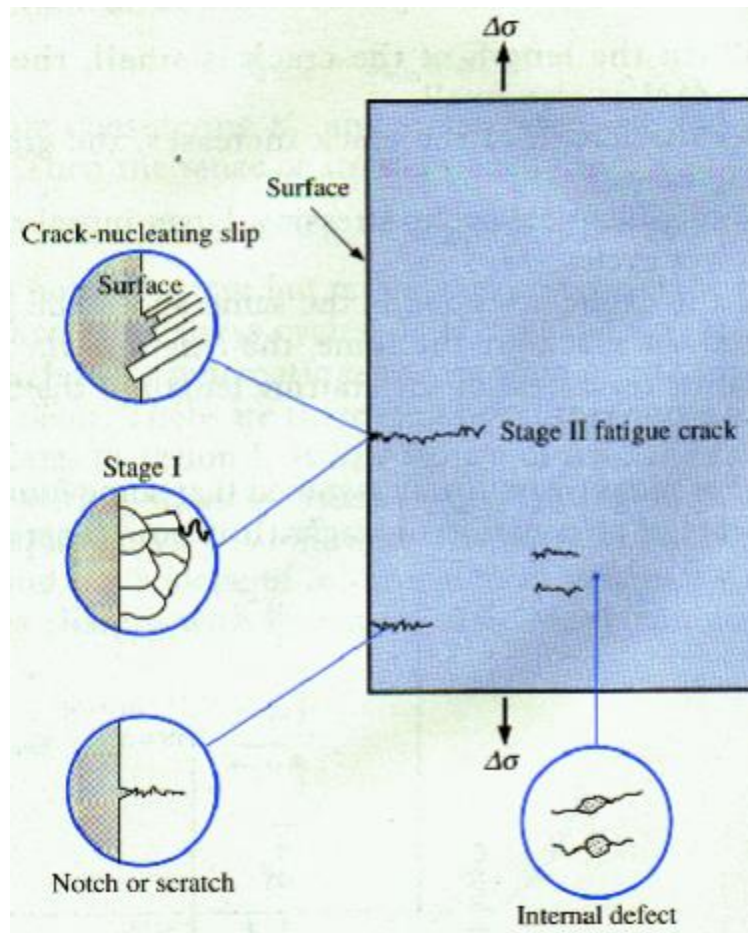
1. Beachmarks, or clamshell marks, are macrofracture features that may be seen in fatigue failures of materials that are used for a period of time, allowed to rest for an equivalent time period and the loaded again as in factory usage.
2. Striations are microfracture features thought to be steps in crack propagation, where the distance depends on the stress range. Beachmarks may contain thousands of striations. Visual Examples of Beachmarks and Striations.



EX/ An example of Beachmarks or "clamshell pattern" associated with stress cycles that vary in magnitude and time as in factory machinery.



EX/ An example of the striations found in fatigue fracture on Microscale level. Each striation is thought to be the advancement of the crack. There may be thousands of striations in a Beachmarks.



The figure above illustrates the various ways in which cracks are initiated and the stages that occur after they start. This is extremely important since these cracks will ultimately lead to failure of the material if not detected and recognized. The material shown is pulled in tension with a cyclic stress in the y or horizontal, direction. Cracks can be initiated by several different causes, the three that will be discussed here are nucleating slip planes, notches, and internal flaws.

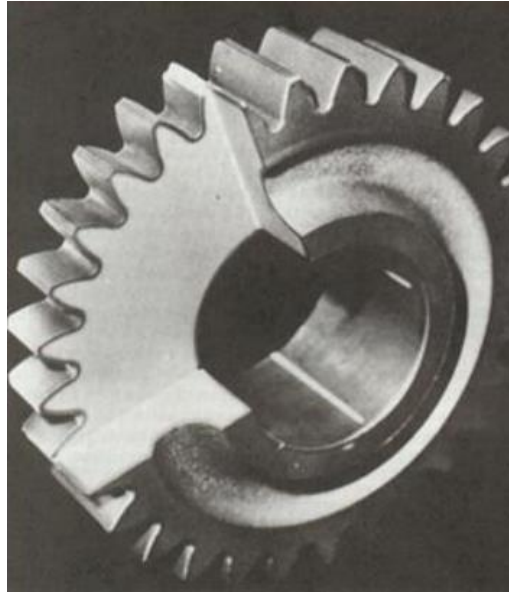
Factors That Affect Fatigue Life and Solutions:

1. Stress level and type
2. Location of the origin
3. Stress concentration level at the origin

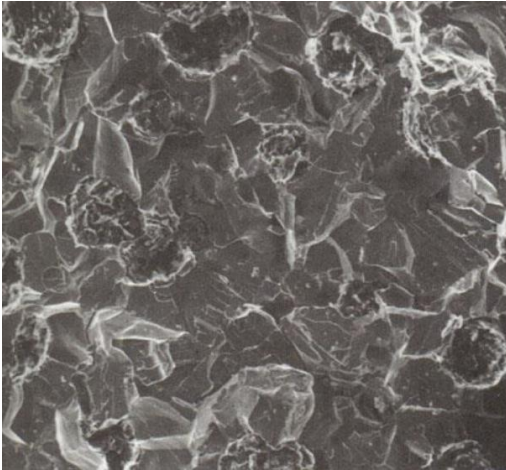
As mentioned previously, scratches and other imperfections on the surface will cause a decrease in the life of a material. Therefore, trying to reduce surface and internal stress concentration sources and imperfections by reducing sharp corners, eliminating unnecessary drilling, and stamping, will enhance the fatigue life of a component.

- Shot peening, and most of all careful fabrication and handling of the material.
- Another Surface treatment is called case hardening, which increases surface hardness and fatigue life. This is achieved by exposing the component to a carbon-rich atmosphere at high temperatures. Carbon diffuses into the material filling interstices and other vacancies in the material, up to 1 mm in depth.

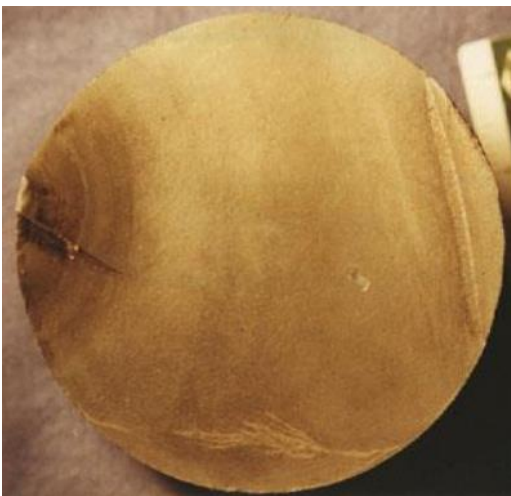
EX/ A case hardened steel gear. Notice the effect of diffusion of Carbon into the material produces a "case" around the gear.



Exposing a material to high temperatures is another cause of fatigue in materials. Thermal expansion and contraction will weaken bonds in a material as well as bonds between two different materials. For example, in space shuttle heat shield tiles, the outer covering of silicon tetraboride (SiB_4) has a different coefficient of thermal expansion than the Carbon-Carbon Composite. Upon re-entry into the earth's atmosphere, this thermal mismatch will cause the protective covering to weaken, and eventually fail with repeated cycles. Another environmental effect on a material is chemical attack, or corrosion. Small pits may form on the surface of the material, similar to the effect etching has when trying to find dislocations.

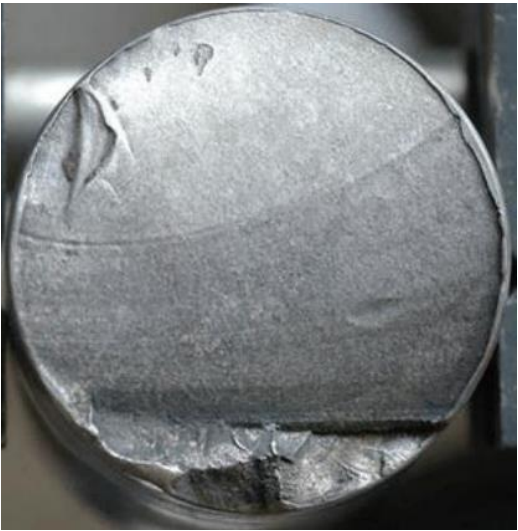
EXAMPLES:

Brittle failure in nodular (circles) cast iron shown by cleavage planes and river patterns.



Origin on the left from shells and from the crack.
Catastrophic zone on the far right.





High Cycle Fatigue (HCF)



Severe stress concentrations with flipped clam shells and crack propagated around catastrophic zone

High applied stresses



Transverse loading as zone is in the middle but not exactly in the center meaning that there is higher stress concentration on the right propagating the cracks faster



Severe stress concentration with severe stresses

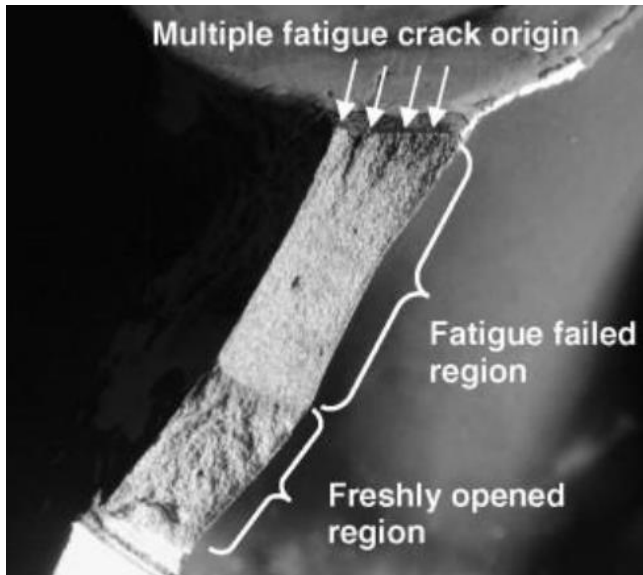


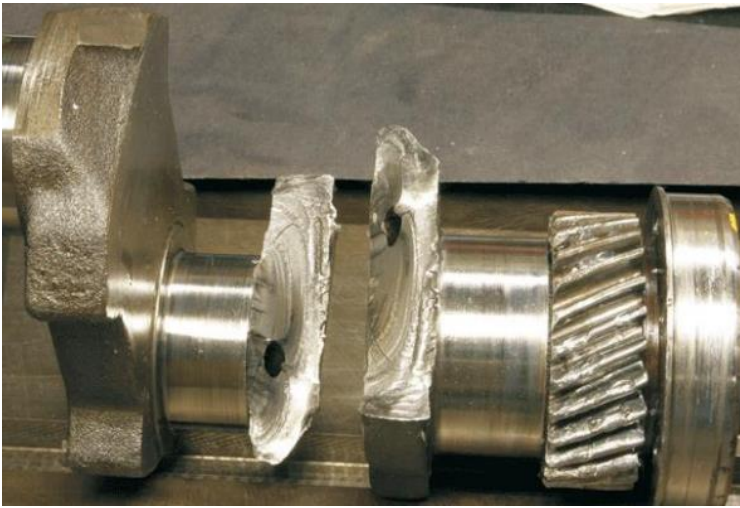
High stress concentrations seen by flat clam shell

Catastrophic zone up and small meaning low applied stresses



Fatigue cracks resulting from torsional stresses showing beach marks and ridges





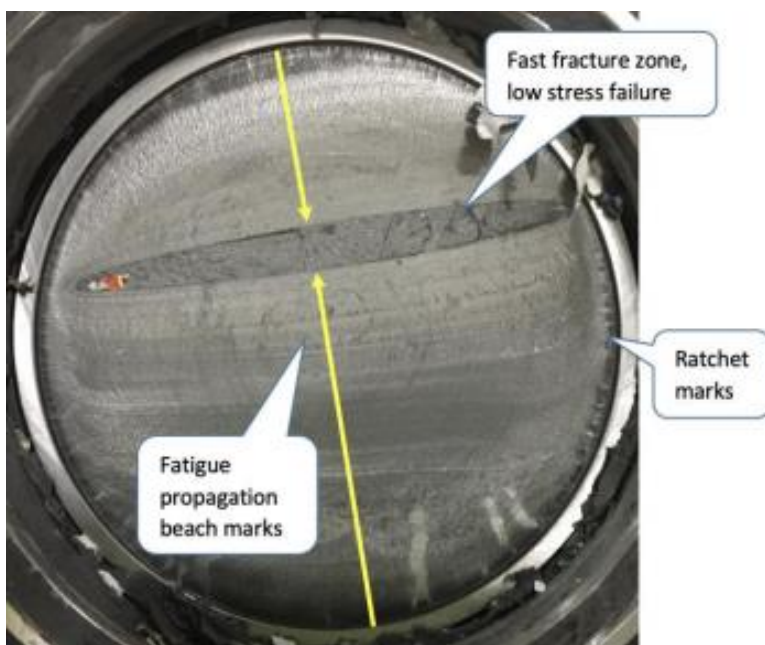
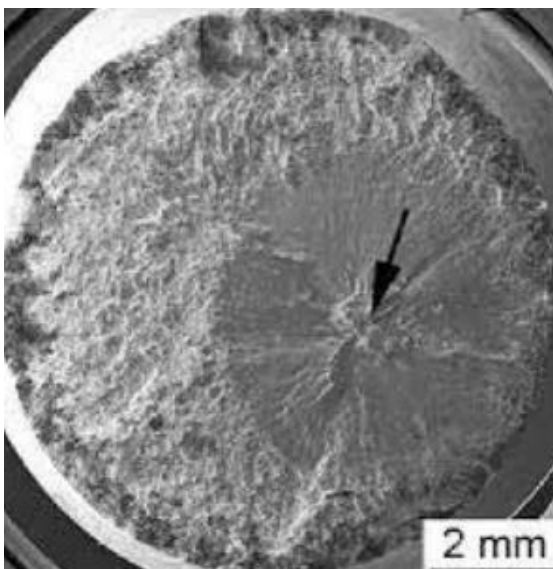
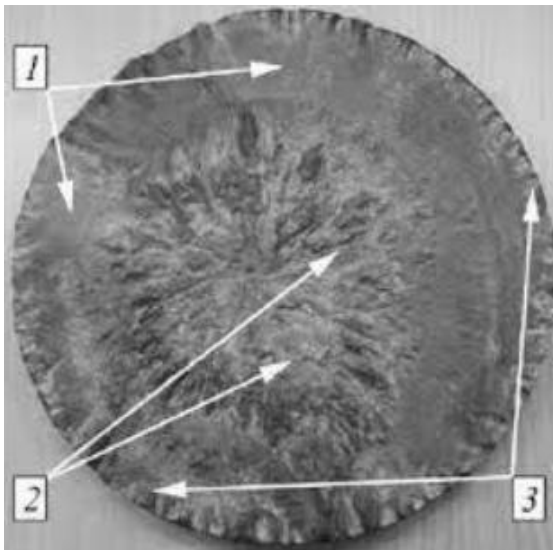
Smooth surface with a distinct 'eye zone' formed by several conspicuous 'beach' marks around a tiny sub-surface defect as the origin (black spot)

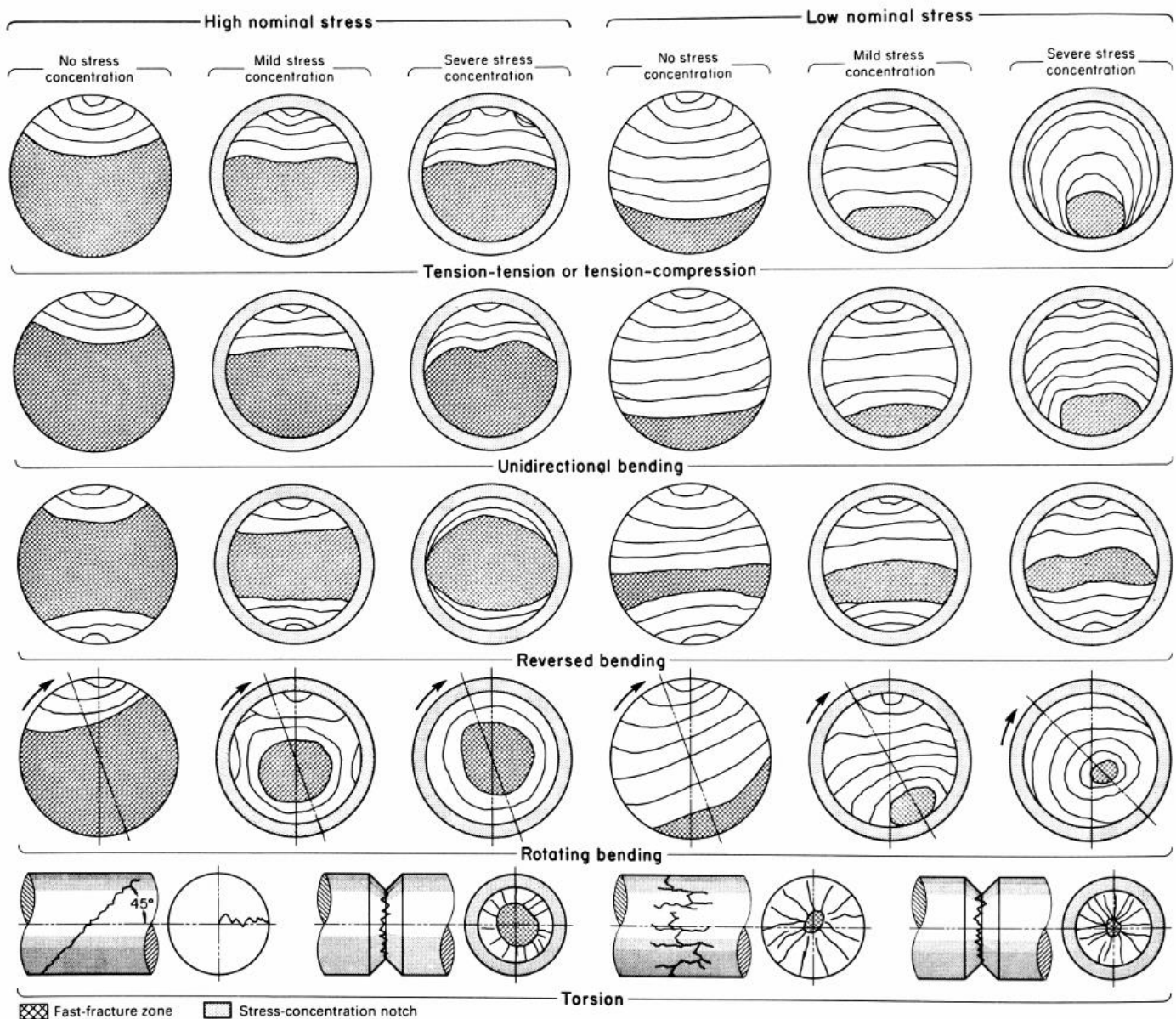


Reversed bending fatigue propagated nearly 100% across the fracture surface



Lighter regions (A) of this fracture are progression marks. The dark region (B) is the area of fast fracture (not fatigue), or the area that was still holding the tensile load just before final fracture. There was a significant percent cross sectional area remaining when fast fracture occurred, indicating the part had a significant clamp load when the part finally fractured.





Schematic representation of fatigue fracture surface marks produced on smooth and notched components with circular cross sections under various loading conditions